Rebuttal Figures and Tables

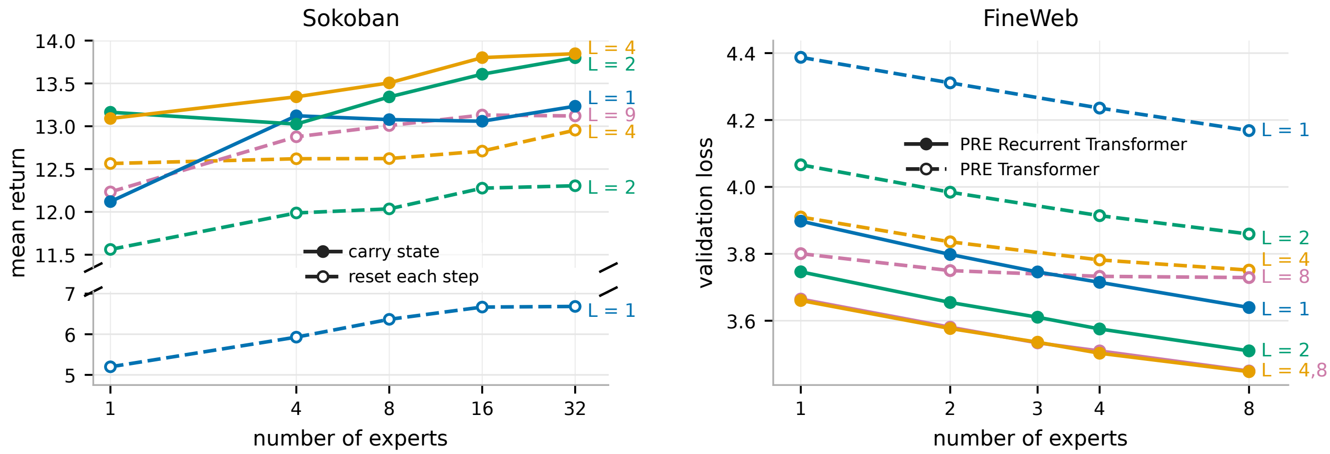


Figure 2: Carried state supplies a serial computation axis. Left: In Sokoban, solid lines carry hidden state across environment steps and dashed lines reset it every step. Resetting state severely hurts shallow updates, while extra within-step depth partially recovers performance; color denotes within-step depth L. **Right**: In FineWeb next-token prediction, the PRE Recurrent Transformer carries latent recurrent state across token steps, while the PRE Transformer recreates latent state from the current segment at each step. The context window size is 128 for both architectures.

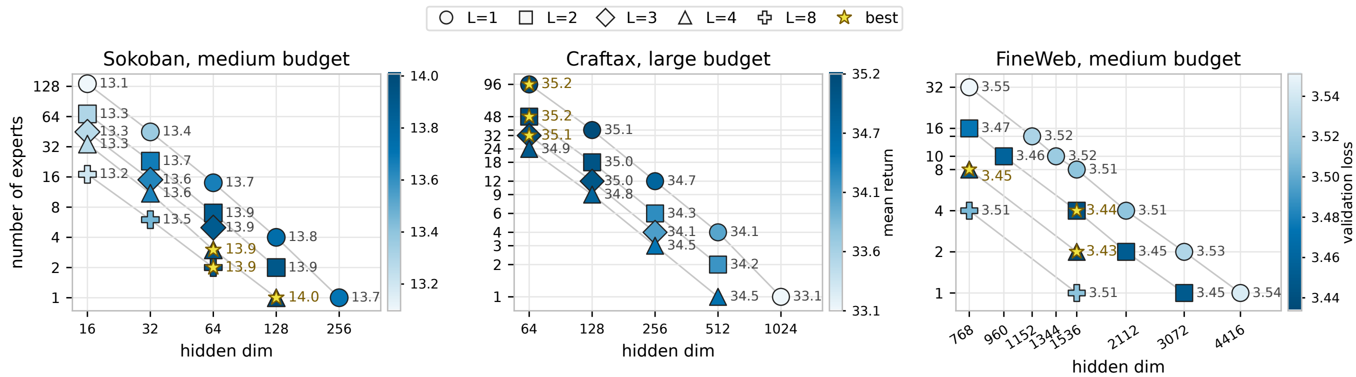


Figure 4: Fixed-compute allocation across domains. The horizontal axis varies expert hidden dimension, the vertical axis varies experts per depth level, and marker shape denotes within-step depth.

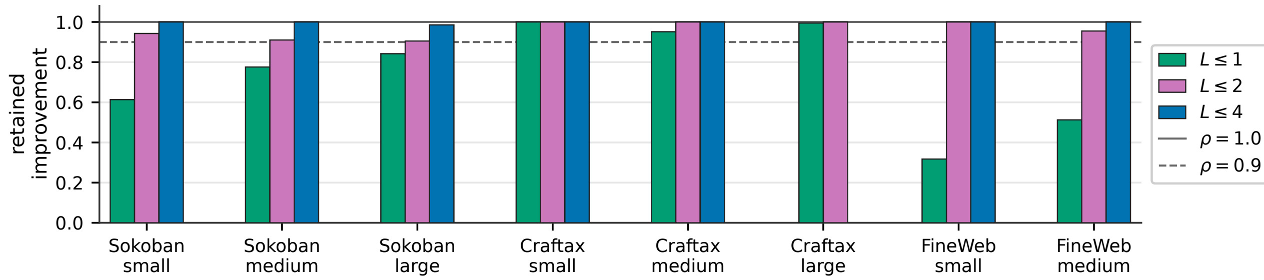


Figure 5: Retained improvement under different depth caps. Each bar reports the retained-improvement ratio $\rho_{\leq k}$ for one matched-budget sweep under depth caps $k \in \{1, 2, 4, 8\}$, with Q defined as episodic return for Sokoban and Craftax and negative validation cross-entropy for FineWeb. Horizontal reference lines mark $\rho = 1.00$, which matches the unrestricted best configuration, and $\rho = 0.90$, which retains 90% of the best improvement over the domain reference.

Table 1: Best observed performance under within-step depth restrictions in the fixed-compute sweeps. Sokoban and Craftax report episodic return, where higher is better. FineWeb next-token prediction reports validation loss, where lower is better. Gap reports the performance loss from the unrestricted optimum: return drop for Sokoban and Craftax, and validation-loss increase for FineWeb. Max L is the largest within-step depth included in that sweep.

Domain	Budget	Max L	Best overall	Best with $L \leq 2$	Gap	Best with $L \leq 1$	Gap
Sokoban	small	8	13.659	13.613	0.046	13.351	0.308
Sokoban	medium	8	14.008	13.905	0.103	13.751	0.257
Sokoban	large	8	14.228	14.099	0.129	14.013	0.215
Craftax	small	8	32.436	32.436	0.000	32.436	0.000
Craftax	medium	8	33.934	33.934	0.000	33.802	0.132
Craftax	large	4	35.200	35.200	0.000	35.179	0.021
FineWeb	small	8	3.481	3.481	0.000	3.559	0.078
FineWeb	medium	8	3.434	3.441	0.007	3.511	0.078

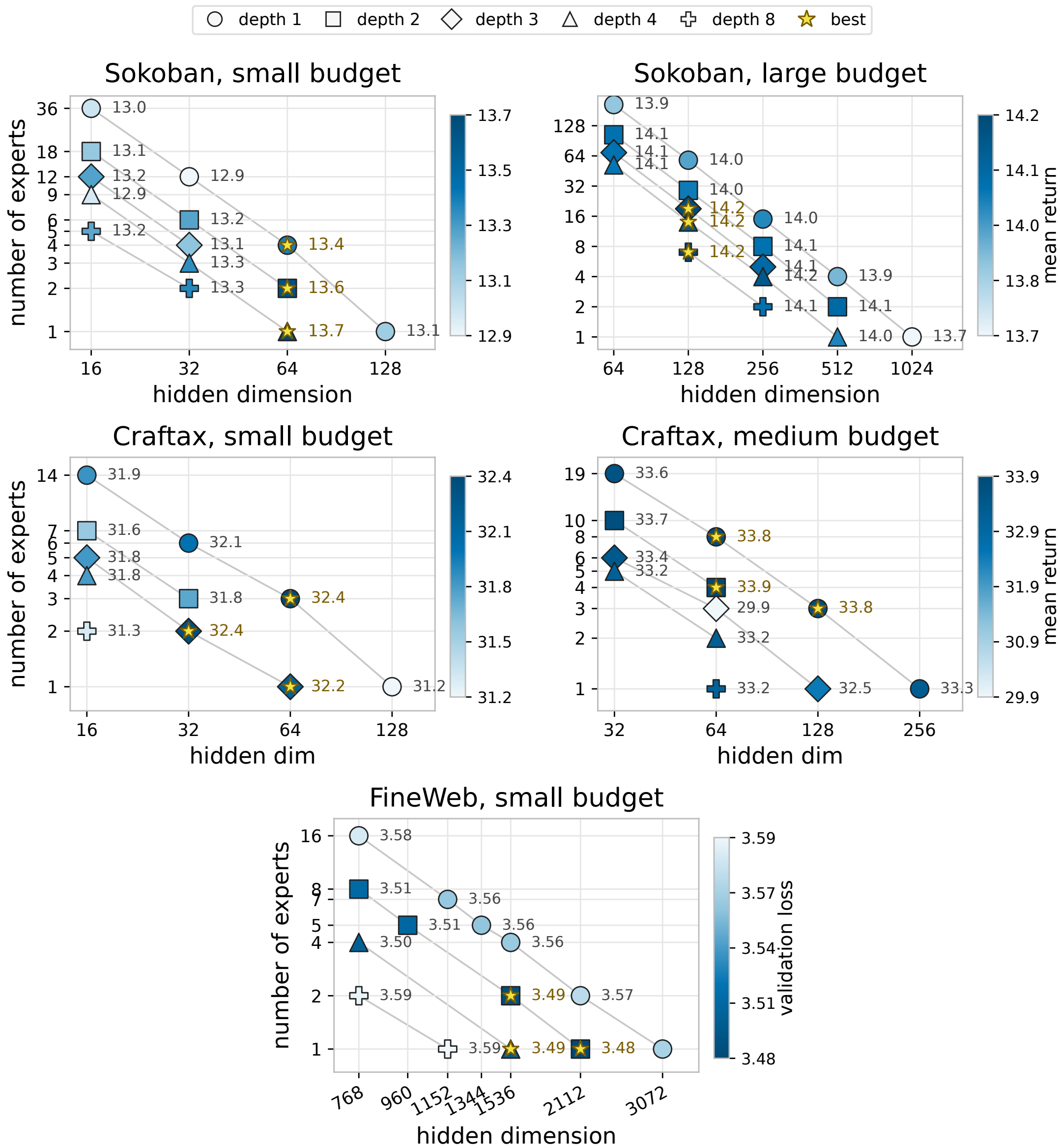


Figure 9: Additional fixed-compute allocation budgets. **Top left**: Sokoban small budget. **Top right**: Sokoban large budget. **Middle left**: Craftax small budget. **Middle right**: Craftax medium budget. **Bottom**: FineWeb small budget. In Sokoban and Craftax, panels vary expert hidden dimension, experts per within-step depth level, and within-step depth; color shows mean return, and only completed or near-completed configurations are shown. In FineWeb, color shows validation loss at the common comparison point of 1.30B consumed tokens, where lower is better.

Table 2: Fixed-compute budgets. Relative work is normalized to the smallest budget within each domain. For Sokoban, $C(d) \propto d(d+16)$; for Craftax, $C(d) \propto d(256+2d)$. Best reports the best observed performance in the matched-budget sweep: return for Sokoban and Craftax, and validation loss for FineWeb.

Domain	Budget	Reference config	Work / small budget	Best
Sokoban	small	$L=1,\ E=1,\ d=128$	1.00	13.659
Sokoban	medium	$L=1,\ E=1,\ d=256$	3.78	14.008
Sokoban	large	$L=1,\ E=1,\ d=1024$	57.78	14.228
Craftax	small	$L=1,\ E=1,\ d=128$	1.00	32.436
Craftax	medium	$L=1,\ E=1,\ d=256$	3.00	33.934
Craftax	large	$L=1,\ E=1,\ d=1024$	36.00	35.200
FineWeb	small	$L=16, E=1,\ d=768$	1.00	3.481
FineWeb	medium	$L=32, E=1,\ d=768$	2.00	3.434

Table 3: Best exact-depth comparison, $L = 4$ versus $L = 2$. For each depth, the configuration with the best observed mean higher-is-better score Q is selected. ΔQ is the observed difference between sample means $\bar{Q}_4 - \bar{Q}_2$, so positive values favor $L = 4$. The p-value is from Welch's t-test of $H_0 : \mathbb{E}[Q_4] \leq \mathbb{E}[Q_2]$ against $H_1 : \mathbb{E}[Q_4] > \mathbb{E}[Q_2]$. n is the number of independent runs per configuration.

Domain	Budget	Best $L = 4$	Best $L = 2$	ΔQ	n	p
Sokoban	small	$E = 1, d = 64$	$E = 2, d = 64$	0.046	3	0.2947
Sokoban	medium	$E = 1, d = 128$	$E = 2, d = 128$	0.102	3	0.0657
Sokoban	large	$E = 14, d = 128$	$E = 2, d = 512$	0.074	3	0.1005
Craftax	small	$E = 4, d = 16$	$E = 3, d = 32$	0.063	3	0.4407
Craftax	medium	$E = 2, d = 64$	$E = 4, d = 64$	-0.690	3	0.9275
Craftax	large	$E = 24, d = 64$	$E = 48, d = 64$	-0.266	3	0.8364

Table 4: Best exact-depth comparison, $L = 2$ versus $L = 1$. For each depth, the configuration with the best observed mean higher-is-better score Q is selected. ΔQ is the observed difference between sample means $\bar{Q}_2 - \bar{Q}_1$, so positive values favor $L = 2$. The p-value is from Welch's t-test of $H_0 : \mathbb{E}[Q_2] \leq \mathbb{E}[Q_1]$ against $H_1 : \mathbb{E}[Q_2] > \mathbb{E}[Q_1]$. n is the number of independent runs per configuration.

Domain	Budget	Best $L = 2$	Best $L = 1$	ΔQ	n	p
Sokoban	small	$E = 2, d = 64$	$E = 4, d = 64$	0.262	3	0.0460
Sokoban	medium	$E = 2, d = 128$	$E = 4, d = 128$	0.154	3	0.0485
Sokoban	large	$E = 2, d = 512$	$E = 15, d = 256$	0.086	3	0.0745
Craftax	small	$E = 3, d = 32$	$E = 3, d = 64$	-0.670	3	0.8968
Craftax	medium	$E = 4, d = 64$	$E = 8, d = 64$	0.132	3	0.3551
Craftax	large	$E = 48, d = 64$	$E = 96, d = 64$	0.021	3	0.4545

Table 5: Best $L = 2$ versus all $L = 1$ configurations in Sokoban. For each small and medium budget, the $L = 2$ configuration with the best observed mean return is compared with every $L = 1$ configuration. Positive ΔQ favors $L = 2$, and p is from the one-sided Welch test defined above. n is the number of independent runs per configuration.

Budget	Best $L = 2$	$L = 1$ configuration	ΔQ	n	p
small	$E = 2, d = 64$	$E = 4, d = 64$	0.262	3	0.0460
small	$E = 2, d = 64$	$E = 1, d = 128$	0.529	3	0.0141
small	$E = 2, d = 64$	$E = 36, d = 16$	0.647	3	0.0029
small	$E = 2, d = 64$	$E = 12, d = 32$	0.749	3	0.0003
medium	$E = 2, d = 128$	$E = 4, d = 128$	0.154	3	0.0485
medium	$E = 2, d = 128$	$E = 1, d = 256$	0.206	3	0.0093
medium	$E = 2, d = 128$	$E = 14, d = 64$	0.248	3	0.0109
medium	$E = 2, d = 128$	$E = 45, d = 32$	0.543	3	0.0004
medium	$E = 2, d = 128$	$E = 136, d = 16$	0.796	3	0.0007

Table 6: Per-seed Sokoban fixed-compute results. These are all points shown for this benchmark in the fixed-compute allocation figures. Mean and standard error are computed across the available seed values. A dash denotes a missing seed or an undefined single-run standard error.

Budget	L	E	d	Seed 1	Seed 2	Seed 3	Mean	SE
small	1	1	128	12.874	13.109	13.269	13.084	0.115
small	1	4	64	13.381	13.174	13.499	13.351	0.095
small	1	12	32	12.755	12.926	12.909	12.864	0.054
small	1	36	16	12.796	13.082	13.020	12.966	0.087
small	2	2	64	13.620	13.700	13.518	13.613	0.053
small	2	6	32	13.223	13.185	13.296	13.235	0.033
small	2	18	16	13.172	13.124	13.022	13.106	0.044
small	3	4	32	12.963	13.223	13.195	13.127	0.082
small	3	12	16	13.155	13.485	13.082	13.241	0.124
small	4	1	64	13.559	13.757	13.660	13.659	0.057
small	4	3	32	13.415	13.370	13.130	13.305	0.089
small	4	9	16	12.703	13.142	12.948	12.931	0.127
small	8	2	32	13.273	13.252	13.324	13.283	0.022
small	8	5	16	13.516	13.325	12.831	13.224	0.204
medium	1	1	256	13.713	13.635	13.750	13.699	0.034
medium	1	4	128	13.672	13.722	13.860	13.751	0.056
medium	1	14	64	13.674	13.652	13.646	13.657	0.008
medium	1	45	32	13.422	13.281	13.384	13.362	0.042
medium	1	136	16	13.097	13.130	13.101	13.110	0.011
medium	2	2	128	13.862	13.868	13.986	13.905	0.040
medium	2	7	64	13.873	13.821	13.890	13.861	0.021
medium	2	23	32	13.656	13.576	13.784	13.672	0.061
medium	2	68	16	13.349	13.436	13.109	13.298	0.098
medium	3	5	64	13.905	13.899	13.838	13.881	0.022
medium	3	15	32	13.781	13.524	13.604	13.637	0.076
medium	3	45	16	13.150	13.451	13.194	13.265	0.094
medium	4	1	128	13.970	14.079	13.974	14.008	0.036
medium	4	3	64	13.833	14.091	13.875	13.933	0.080
medium	4	11	32	13.624	13.596	13.726	13.649	0.040
medium	4	34	16	13.039	13.434	13.324	13.266	0.118
medium	8	2	64	14.012	13.900	13.884	13.932	0.040
medium	8	6	32	13.644	13.339	13.385	13.456	0.095
medium	8	17	16	13.074	13.384	13.148	13.202	0.093
large	1	1	1024	13.783	13.573	13.790	13.715	0.071
large	1	4	512	13.907	13.862	13.971	13.914	0.032
large	1	15	256	14.004	13.999	14.035	14.013	0.011
large	1	58	128	13.990	13.852	14.027	13.956	0.053
large	1	208	64	13.809	13.931	13.891	13.877	0.036
large	2	2	512	14.051	14.070	14.176	14.099	0.039
large	2	8	256	14.141	13.899	14.148	14.063	0.082
large	2	29	128	14.048	13.974	14.102	14.041	0.037
large	2	104	64	14.063	14.067	14.079	14.070	0.005
large	3	5	256	14.142	14.057	14.090	14.096	0.025
large	3	19	128	14.232	14.168	14.226	14.209	0.020
large	3	69	64	13.988	14.103	14.104	14.065	0.039
large	4	1	512	13.957	14.104	14.007	14.022	0.043
large	4	4	256	14.194	14.089	14.208	14.164	0.037
large	4	14	128	14.125	14.214	14.179	14.173	0.026
large	4	52	64	14.042	14.035	14.081	14.053	0.014
large	8	2	256	14.143	14.039	14.010	14.064	0.040
large	8	7	128	14.145	14.281	14.259	14.228	0.042

Table 7: Per-seed Craftax fixed-compute results. These are all points shown for this benchmark in the fixed-compute allocation figures. Mean and standard error are computed across the available seed values. A dash denotes a missing seed or an undefined single-run standard error.

Budget	L	E	d	Seed 1	Seed 2	Seed 3	Mean	SE
small	1	1	128	29.480	31.632	32.538	31.216	0.907
small	1	3	64	32.367	32.800	32.142	32.436	0.193
small	1	6	32	31.592	32.166	32.393	32.050	0.238
small	1	14	16	31.771	31.964	31.866	31.867	0.056
small	2	3	32	32.132	32.141	31.026	31.766	0.370
small	2	7	16	31.171	31.600	31.977	31.582	0.233
small	3	1	64	32.755	31.168	32.701	32.208	0.520
small	3	2	32	32.198	32.390	32.557	32.381	0.104
small	3	5	16	32.047	31.144	32.337	31.843	0.359
small	4	4	16	31.927	31.769	31.791	31.829	0.049
small	8	2	16	31.191	31.510	31.252	31.318	0.098
medium	1	1	256	33.611	33.255	33.034	33.300	0.168
medium	1	3	128	33.481	33.792	34.011	33.761	0.154
medium	1	8	64	33.573	33.952	33.881	33.802	0.116
medium	1	19	32	33.299	33.933	33.635	33.622	0.183
medium	2	4	64	34.372	34.060	33.370	33.934	0.296
medium	2	10	32	33.556	33.955	33.658	33.723	0.120
medium	3	1	128	33.713	30.396	33.326	32.478	1.047
medium	3	3	64	34.636	21.646	33.337	29.873	4.130
medium	3	6	32	33.660	33.012	33.606	33.426	0.207
medium	4	2	64	32.895	33.153	33.685	33.244	0.233
medium	4	5	32	33.349	32.919	33.252	33.173	0.130
medium	8	1	64	33.488	33.150	32.814	33.150	0.194
large	1	1	1024	33.044	34.154	31.969	33.056	0.631
large	1	4	512	33.405	34.443	34.306	34.051	0.326
large	1	12	256	34.329	34.919	34.959	34.736	0.204
large	1	36	128	35.190	35.049	35.172	35.137	0.044
large	1	96	64	35.228	35.285	35.024	35.179	0.079
large	2	2	512	34.128	34.553	33.895	34.192	0.193
large	2	6	256	34.223	33.614	34.942	34.260	0.384
large	2	18	128	34.646	34.683	35.735	35.021	0.357
large	2	48	64	35.490	35.131	34.980	35.200	0.151
large	3	4	256	33.728	34.438	34.156	34.107	0.206
large	3	12	128	34.878	35.186	34.882	34.982	0.102
large	3	32	64	34.726	35.700	34.995	35.140	0.291
large	4	1	512	34.670	34.525	34.392	34.529	0.080
large	4	3	256	34.273	34.422	34.738	34.477	0.137
large	4	9	128	34.966	34.542	34.787	34.765	0.123
large	4	24	64	35.020	34.583	35.199	34.934	0.183

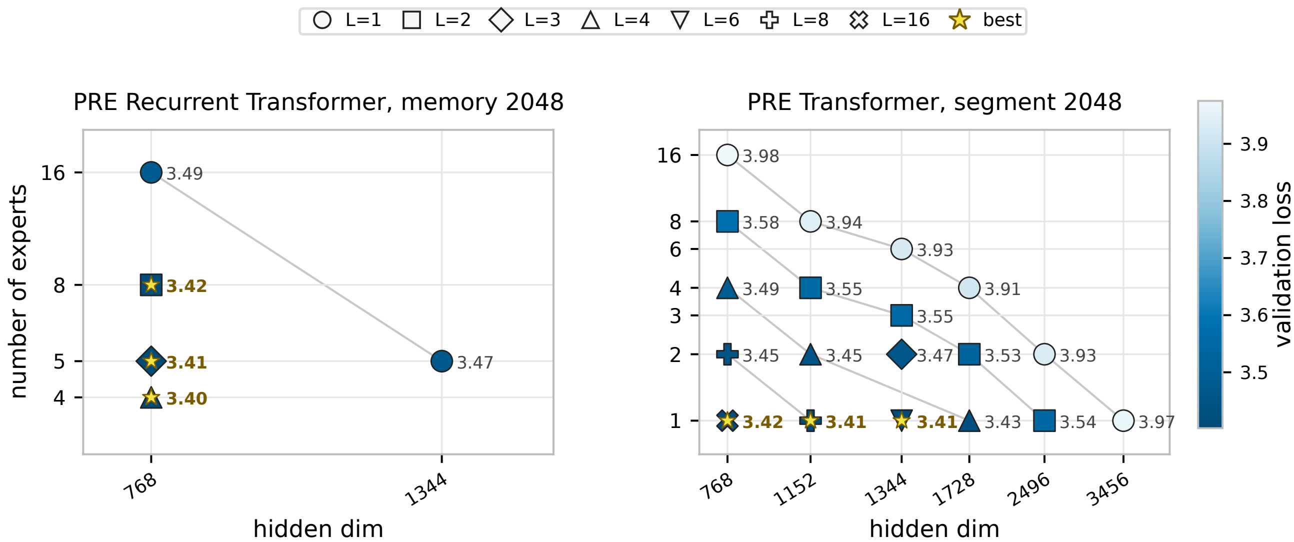


Figure 12: FineWeb budget-16 comparison at long context. **Left**: completed finite PRE Recurrent Transformer runs with recurrent memory length 2048 and segment length of 64. **Right**: the corresponding non-recurrent GPT-style PRE Transformer fixed-budget sweep with training segment length 2048. Color and point labels show validation loss at the latest common logged validation point within each panel, where lower is better; marker shape denotes within-step depth L, and stars mark the best configurations within each panel. The PRE Recurrent Transformer panel contains the subset of memory-2048 configurations that completed with finite validation loss, while the GPT-style PRE Transformer panel contains the completed segment-2048 sweep.

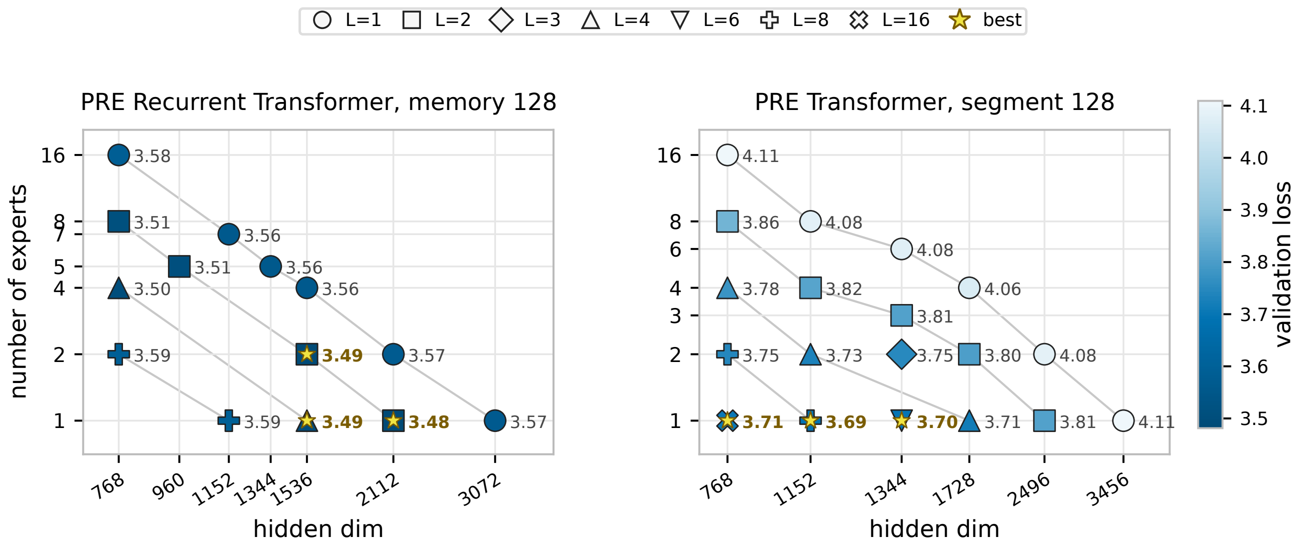
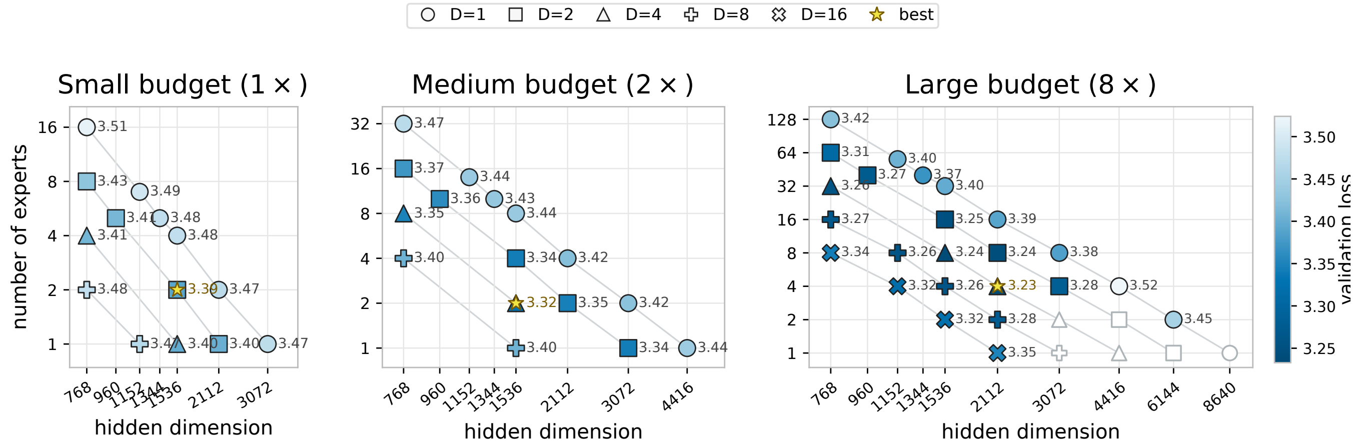


Figure 13: FineWeb budget-16 comparison at context length 128. **Left**: the PRE Recurrent Transformer small-budget allocation map, repeated here for direct comparison. The segment length is 64 and the memory length is 128. **Right**: a non-recurrent GPT-style PRE Transformer sweep with training segment length 128 and the same nominal budget. Color and point labels show validation loss at the latest common logged validation point within each panel, where lower is better; marker shape denotes within-step depth L, and stars mark the best configurations within each panel.



Async-paper Figure 2. FineWeb fixed-recurrent-compute allocation across three budgets. Left: small budget, with all 14 configurations complete. **Middle:** medium budget, twice the small-sweep recurrent compute, with all 16 configurations complete. **Right:** large budget, eight times the small-sweep recurrent compute, with 24 of 30 configurations complete, including completed low-expert and depth-16 extensions. All panels share one validation-loss color scale. Colored points and labels report the final validation checkpoint near 1.295B consumed tokens; all but one use W&B step 4,940, while the lower-PDB $D = 1, E = 2, W = 6144$ run uses its nearest checkpoint at step 4,953. Hollow points mark planned configurations that had not completed when this snapshot was generated. Circles, squares, triangles, filled-plus markers, and crosses denote $D = 1, 2, 4, 8, 16$, respectively; a star marks the best completed configuration in each panel. Lower loss is better. All runs use sequence length 128, memory length 128, a 262,144-token optimizer batch, sandwich RMSNorm, and a persistent K/V cache shared across depth. The large panel remains interim while its hollow configurations finish.

Table 10: Four-H100 median steady-state decode latency in milliseconds. GPT-16 uses cached autoregressive decoding with a full sliding-window key-value cache; PRE uses segment-local recurrent decoding with depth-shared recurrent key-value memory. B is the global batch and $B/4$ is the per-GPU batch.

Model	$C = 128, B = 16$	$C = 2048, B = 16$
GPT-16	0.636	0.760
PRE $D = 1, E = 16$	0.133	0.209
PRE $D = 2, E = 8$	0.498	0.645
PRE $D = 4, E = 4$	0.694	0.943
PRE $D = 8, E = 2$	0.943	1.447
PRE $D = 16, E = 1$	1.465	2.454

Table 11: Four-H100 FineWeb training throughput in tokens/s. Each panel fixes token batch CB and reports segment length C and global sequence batch B. Values are medians over steps 2–5; startup, compilation, and autotuning steps 0–1 are excluded. OOM denotes measured memory exhaustion. All displayed values use one microbatch per optimizer update.

131k effective tokens per optimizer update						
Model	$C = 32$ $B = 4{,}096$	$C = 64$ $B = 2{,}048$	$C = 128$ $B = 1{,}024$	$C = 256$ $B = 512$	$C = 1024$ $B = 128$	$C = 2048$ $B = 64$
GPT-16	1,004.7k	1,056.6k	1,098.2k	1,113.8k	1,066.1k	1,021.9k
PRE $D = 1, E = 16$	471.3k	417.4k	327.2k	218.7k	67.0k	OOM
PRE $D = 2, E = 8$	229.8k	182.0k	128.1k	79.2k	22.0k	OOM
PRE $D = 4, E = 4$	169.0k	134.6k	96.7k	60.6k	17.1k	OOM
PRE $D = 8, E = 2$	113.8k	88.5k	63.2k	39.4k	11.7k	OOM
PRE $D = 16, E = 1$	68.7k	54.1k	37.7k	24.3k	7.5k	OOM

262k effective tokens per optimizer update						
Model	$C = 32$ $B = 8{,}192$	$C = 64$ $B = 4{,}096$	$C = 128$ $B = 2{,}048$	$C = 256$ $B = 1{,}024$	$C = 1024$ $B = 256$	$C = 2048$ $B = 128$
GPT-16	1,115.5k	1,205.6k	1,238.8k	1,245.7k	1,168.6k	1,125.4k
PRE $D = 1, E = 16$	511.1k	486.0k	419.3k	319.4k	OOM	OOM
PRE $D = 2, E = 8$	263.7k	228.7k	179.4k	121.7k	OOM	OOM
PRE $D = 4, E = 4$	193.9k	166.6k	131.0k	90.0k	OOM	OOM
PRE $D = 8, E = 2$	128.9k	110.2k	84.8k	57.9k	OOM	OOM
PRE $D = 16, E = 1$	77.9k	66.3k	50.5k	34.1k	OOM	OOM

524k effective tokens per optimizer update						
Model	$C = 32$ $B = 16{,}384$	$C = 64$ $B = 8{,}192$	$C = 128$ $B = 4{,}096$	$C = 256$ $B = 2{,}048$	$C = 1024$ $B = 512$	$C = 2048$ $B = 256$
PRE $D = 1, E = 16$	543.6k	517.3k	477.9k	377.4k	OOM	OOM
PRE $D = 2, E = 8$	285.2k	260.2k	219.6k	157.1k	OOM	OOM
PRE $D = 4, E = 4$	208.2k	188.9k	158.9k	112.0k	OOM	OOM
PRE $D = 8, E = 2$	137.6k	124.0k	103.8k	OOM	OOM	OOM
PRE $D = 16, E = 1$	83.7k	74.4k	60.4k	OOM	OOM	OOM